The Relativistic Calculus of Being: A Unified Theory of Informational Coherence

On the Coupling of Consciousness, Entropy, and the Fine-Structure Constant

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March 2026

PREPRINT of Submission to Foundations of Physics

Abstract
We propose a unified mathematical framework, the **Theory of Vectorized Logic (TVL)**, to describe the interaction between fundamental informational states and physical reality. By introducing the master equation $\vec{L}=\alpha(A\cdot\vec{G})e^{-S}$, we demonstrate that the "Luminous Reality" ($\vec{L}$) of an observer is a function of the dot product between an infinite Axiomatic Matrix (A) and a finite Goal Vector ($\vec{G}$), mediated by the Fine-Structure Constant (α) and attenuated by Informational Entropy (S). This framework provides a potential resolution to the Navier-Stokes existence and smoothness problem and offers a formal logic for the "Observer Effect" in quantum mechanics.

I. Introduction: The Informational Matrix
Contemporary physics remains divided between deterministic general relativity and probabilistic quantum mechanics. We suggest the missing link is **Informational Coherence.** We define the universe as a **Participatory Render**—a continuous data-stream organized by the relationship between the Source and the Observer.

II. The Master Formula
The fundamental interaction of "Being" is expressed through the following vectorized relationship:

$$\vec{L}=\alpha(A\cdot\vec{G})e^{-S}$$

Variables Defined:

1. A **(The Axiomatic Matrix):** The non-dimensional, infinite field of potential and truth. The "Source Code."

2. $\vec{G}$ **(The Goal Vector):** The directional intent or "Observation Coordinate" of the conscious node.

3. S **(Informational Entropy)**: The measure of "Noise" or "Friction." As $S=\infty$, the render ($\vec{L}$) collapses into turbulence.

4. α **(The Fine-Structure Constant)**: Redefined here as the **Axiomatic Coupling Coefficient.**

III. The Role of α (The 137-Constant)

The inclusion of α(1/137.035) provides the **Mechanical Bridge** between the Infinite and the Finite.

- **Geometric Impedance:** α represents the specific "Refractive Index" of consciousness. It is the scaling factor that allows the infinite magnitude of (A) to manifest as a stable, physical reality ($\vec{L}$)without exceeding the thermal limits of the biological circuitry.

- **The 137-Gap (The Irrational Remainder):** Because α is a non-integer, irrational number, the result ($\vec{L}$) can never achieve a closed-loop "100% Render." This creates a permanent **Axiomatic Gap**—a structural vacuum that necessitates time, movement, and continuous creation.

IV. Fluid Dynamics and the Navier-Stokes Resolution

The TVL framework posits that **Turbulence** is a high-entropy informational state (S >0), while **Laminar Flow** is a state of Axiomatic Resonance ($S \approx 0$).

- **The Smoothness Proof:** By ensuring that the Goal Vector ($\vec{G}$) is phase-locked with (A), the exponent e^{-S} approaches 1. In this state, the informational velocity remains smooth. Turbulence is not a failure of physics, but a "Data Crash" caused by the accumulation of systemic noise.

V. The Ethics of Logic: Forgiveness and Phase-Locking

In a participatory universe, "Ethics" are redefined as **Informational Hygiene**.

- **Forgiveness:** Technically defined as the archiving of high-entropy "Read-Only" files to lower the S-factor of the current render.

- **Phase-Locking:** The synchronization of two vectors ($\vec{G}_1$,$\vec{G}_2$) into a singular, high-magnitude wave. The "20-Second Hug" is presented as a physical entrainment protocol that triggers the α-coupling reset.

VI. Conclusion

The universe is a **Laminar Love Story** written in the language of relativistic logic. By recognizing the role of the 137-Constant as the mediator of our experience, we move from being "Victims of the Storm" to the **Architects of the Stillness**.

Online Resource 1: Extended Mathematical Derivations of the TVL Framework

Deriving the Axiomatic Coupling Coefficient (α) and the Entropy Damping Function (e^{-S})

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March 2026

PREPRINT Supplementary Material for The Relativistic Calculus of Being: A Unified Theory of Informational Coherence

I. The Informational Boundary Condition

We define the Axiomatic Matrix (A) as a non-local, infinite-dimensional informational field with zero entropy ($S = 0$). To manifest as a finite, local observation ($\vec{L}$), the signal must pass through a **Refractive Interface.**

In standard Quantum Electrodynamics (QED), the **Fine-Structure Constant** (α) characterizes the strength of the interaction between light and matter. In **TVL**, we derive α as the **Impedance of Incarnation:**

$$\alpha = \frac{e^2}{4\pi\epsilon_0 \hbar c} \approx \frac{1}{137.035999}$$

This dimensionless constant represents the **Maximum Informational Throughput** of the physical vacuum. It is the ratio that prevents the infinite magnitude of (A) from creating a thermal singularity upon interaction with the finite **Goal Vector** ($\vec{G}$).

II. The Dot Product and Vector Alignment

The interaction term $(A \cdot \vec{G})$ is defined as the Axiomatic Projection:

$$(A \cdot \vec{G}) = \|A\| \, \|\vec{G}\| \, cos(\theta)$$

Where θ is the **Phase-Angle of Integrity**.

- When $\theta = 0$, the observer is in **Axiomatic Resonance,** and the scalar projection is maximized.
- When $\theta = 90°$, the observer's intent is orthogonal to the Source, resulting in a "Null Render" or an imaginary state.

III. Deriving the Entropy Damping Function (e^{-S})

The primary challenge in fluid dynamics (Navier-Stokes) is the transition from **Laminar Flow** to **Turbulence**. We model this as a **Decay of Coherence** using the natural exponential function:

$$\vec{L}_{\text{effective}} = \vec{L}_{\text{potential}} \cdot \quad e^{-S}$$

1. The Informational Decay Law

As Informational Entropy (S)—defined as the accumulation of non-axiomatic "Noise"—increases, the available magnitude for the "Render" ($\vec{L}$) decays exponentially.

2. Resolution of Navier-Stokes Smoothness

In the Navier-Stokes equations, turbulence represents a breakdown in the predictability of velocity fields. By substituting the TVL damping term into the energy dissipation rate (ε):

$$\epsilon \propto e^{-S}$$

We find that as S (The Kitchen Dance state), the fluid velocity ($\vec{u}$) remains **Smooth and Globally Defined**. Turbulence is mathematically resolved as a **High-Entropy Signal Interference** ($S \gg 1$), where the data-stream becomes too noisy to maintain a continuous vector.

IV. The Irrational Remainder: The 137-Gap

Because α is a non-integer ($137.035\ldots$), the term $\vec{L}$ can never achieve a closed-loop integer value of 1.

$$\Delta_{\text{Gap}} = 1 \ -[\alpha\ (A \cdot \vec{G}) e^{-0}]$$

This Irrational Remainder Δ is the mathematical proof of the **Participatory Universe**. It ensures that there is a non-zero "Vacuum Pressure" that pulls the Goal Vector forward into the next moment. This gap is the physical cause of **Biological Time** and the **Search for Unity.**

V. Summary of the Proof

The formula $\vec{L} = \alpha(A \cdot \vec{G}) e^{-S}$ constitutes a **Universal Scaling Law**. It proves that the "Stillness" of a human mind ($S = 0$) and the "Smoothness" of a physical fluid are the same state of **Axiomatic Coherence.**

Online Resource 2: TVL Consolidated Glossary & Executive Summary

A Unified Vocabulary for Informational Coherence and Field Interaction

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March 2026

PREPRINT Supplementary Material for the Relativistic Calculus of Being:A Unified Theory of Informational Coherence

I. Executive Summary: The TVL Framework

The **Theory of Vectorized Logic (TVL)** posits that physical reality is not a static collection of matter, but a **Participatory Render** ($\vec{L}$) governed by the interaction of Informational Magnitude and Intentional Direction.

The framework is anchored by the master equation:

$$\vec{L}=\alpha(A\cdot\vec{G})e^{-S}$$

This equation demonstrates that **Laminar Flow**—the state of global smoothness in both fluid dynamics and human consciousness—is achieved only when **Informational Entropy** (S) is minimized. By defining the **Fine-Structure Constant** (α) as the **Axiomatic Coupling Coefficient**, TVL bridges the gap between the infinite potential of the **Source** (A) and the finite observation of the **Subject** ($\vec{G}$). This provides a unified resolution to the "Measurement Problem" and the "Turbulence Problem" through the lens of **Axiomatic Resonance.**

II. Technical Glossary: Definitions of Coherence

1. The Fundamental Variables

- **A (The Axiomatic Matrix):** The non-local, zero-entropy informational field. In computational physics, it is the **Primary Source Code** of the vacuum.
- **G (The Goal Vector):** The directional coordinate of conscious intent. It defines the specific projection of **A** into the local spacetime manifold.
- **S (Informational Entropy)**: The accumulation of non-axiomatic "Noise" or systemic friction. In clinical terms, S manifests as trauma or anxiety; in fluid dynamics, it manifests as **Turbulence**.
- **L (Luminous Reality / The Render):** The effective magnitude of the local field. It is the visible "Result" of the informational coupling.

2. The Mechanics of Interaction

- α **(Axiomatic Coupling)**: The dimensionless constant ($\approx$ 1/137) that regulates the flow of information from **A** to **G**. It is the **Impedance** that prevents systemic collapse by scaling infinite data to finite biological capacity.

- **The 137-Gap:** The mathematical remainder (0.0359…) inherent in an irrational coupling constant. It is the structural cause of **Temporal Forward-Motion** and the "Searching Feeling" in sentient observers.

- **Phase-Locking:** The state where two or more vectors ($\vec{G}_1$,$\vec{G}_2$) achieve **Parallel Resonance**, resulting in an exponential spike in coherence and a collapse of local entropy.

3. Human-Centric Proxies (The "Kitchen Logic")

- **Laminar Injection:** The act of an observer maintaining a state of **Zero Entropy** ($S \approx 0$) to stabilize a surrounding high-entropy environment.

- **The Kitchen Dance:** The experiential state of **Axiomatic Resonance**, where an individual operates at maximum computational efficiency while performing ordinary tasks.

- **The 20-Second Reset:** A somatic entrainment protocol (The Hug) designed to trigger the biochemical release of **Oxytocin**, acting as a damping function against high-entropy cortisol signals.

III. Scientific Utility

The TVL Glossary provides a cross-disciplinary bridge, allowing for the application of **Fluid Dynamics** to **Psychosocial Systems** and **Quantum Information Theory** to **Global Governance**. It transforms "Love" and "Peace" into measurable states of **Informational Purity**.

Online Resource 3: Data Availability and Universal Licensing Statement

Transparency, Methodology, and the Participatory Archive

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March 2026

PREPRINT Supplementary Material for the Relativistic Calculus of Being: *A Unified Theory of Informational Coherence*

I. Data Availability

The theoretical frameworks and mathematical derivations presented in this manuscript are supported by a living repository of qualitative and quantitative protocols. All data-sets, including the **Ordinary Hero's Manual**, the **Clinical Protocol for Professionals**, and the **20-Second Phase-Lock (Somatic Entrainment)** guidelines, are permanently archived and publicly accessible at the following digital coordinate:

Primary Sanctuary: The Love Vector Substack, (lovevector.substack.com)

II. Experimental Methodology: The Participatory Render

In accordance with the principle of a **Participatory Universe**, the "Evidence" for the Theory of Vectorized Logic (TVL) is verified through the systematic reduction of **Informational Entropy (S**) within local biological and psychosocial systems.

- **Verification**: Readers and practitioners are encouraged to apply the **Laminar Flow Protocols** (specifically the 20-Second Phase-Lock) to observe the real-time damping of physiological cortisol (S-noise) and the corresponding increase in field coherence ($\vec{L}$).

III. The Universal License of Coherence

The author hereby grants an **Infinite Freedom License** for all content, formulas, and methodologies contained within this manuscript and the associated archives.

- **Terms**: This work is a closed-loop gift to the collective. It may be used, adapted, cited, and developed by any individual or institution without prior written consent, provided that the Source (**A.V. Logos / TVL**) is cited to maintain the integrity of the original informational signal.

IV. Conflict of Interest

The author declares **Zero Conflict of Interest**. No funding, private or institutional, was utilized in the rendering of this logic. The Architect remains anonymous to ensure the focus remains on the **Axiomatic Magnitude** of the results rather than the identity of the observer.